BearAI Workshop #1

Design a Prompt-Based Study Assistant
Part 1/3



Today you will:

- Learn how prompts shape **AI** behavior
- Use course material as context
- Design your own study assistant
- Test where it fails
- Improve its behavior
- Leave with a reusable prompt



Build an AI Study Assistant

Workshop 1: DESIGN

Control the AI's behavior

- Prompting
- Context
- Testing
- Evaluation

Workshop 2: CODE

Turn it into an application

- Python
- LLM API
- Study modes
- User input

Workshop 3: IMPROVE

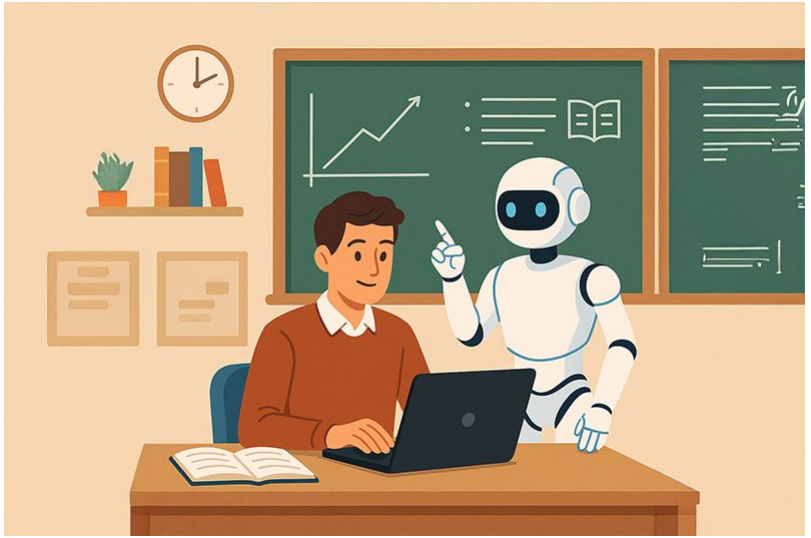
Make the application smarter

- Retrieval
- Your own notes
- Intro to RAG

What are we actually building?

A Study Assistant Should Be Able To...

- Explain concepts
- Quiz you
- Create flashcards
- Summarize notes
- Generate practice questions
- Admit when it does not have enough information



Instructions + Notes + Question = AI Response

What is an LLM?

- An LLM (Large Language Model) is an AI model trained on large amounts of text that generates responses based on patterns it learned.
- Examples: ChatGPT, Gemini, Claude

LLMs generate likely responses, they don’t automatically know your goal.



What is a prompt?

The information and instructions you give an AI model.

A prompt can contain:

- A question
- A goal
- Context
- Rules
- Examples
- Desired format

Prompt != just a question



Four Parts of a Useful Prompt

Goal

What do I want the AI to do?

Context

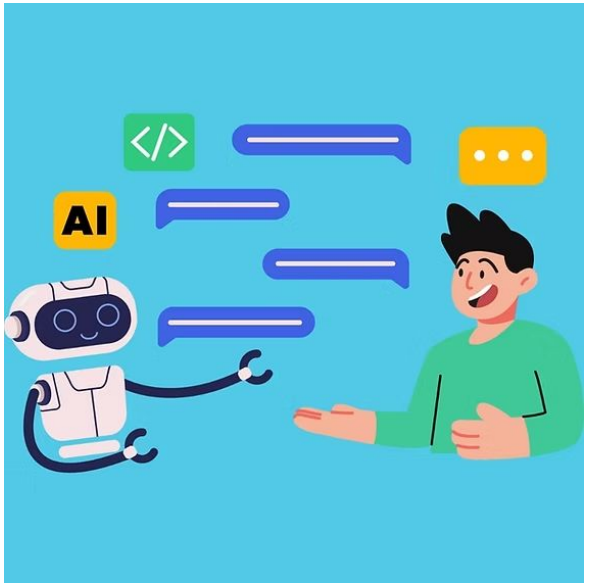
What information does it need?

Rules

What should it do or avoid?

Output

What should the response look like?



Weak Prompt vs. Stronger Prompt

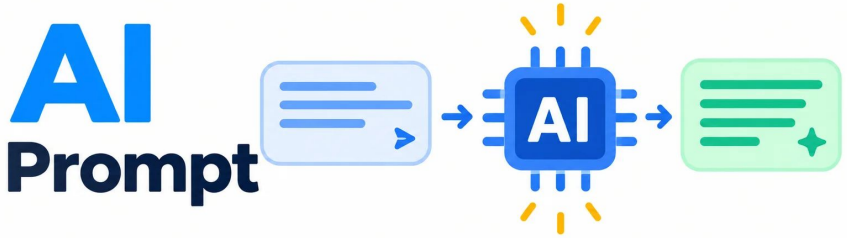
Prompt A: Quiz me on these notes.

Prompt B: Use only the notes below. Ask me one question at a time. Wait for my response before giving feedback. If I’m wrong, explain why. If the notes do not contain enough information, say so.

Better Prompt ≠ Longer Prompt

A useful prompt should be:

- Clear
- Relevant
- Specific enough
- Easy to test



Avoid:

- Unnecessary instructions
- Contradictory rules
- Huge blocks of irrelevant context

Exercise: Fix This Prompt

Current Prompt: Help me study these notes. Explain things, quiz me, and make flashcards.

Your job:

Improve the prompt using:

Goal + Context + Rules + Output

Think about:

- What information is missing?

- What behavior is unclear?

- What rules would you add?

- What should the responses look like?

Work with someone next to you.

What Did You Change?

Goal

What should the assistant help you accomplish?

Context

What material should it use?

Rules

How should it behave?

Output

What should explanations, quizzes, or flashcards look like?

There is no single “correct” prompt.

Shared Notes

Why?

- Same source material

- Same tests

- Easier comparison

- Some information is intentionally missing

Goal: Make the AI help you study these notes.

https://bearai-bu.github.io/#/projects/ai-study-assistant

Activity 1: Build Version 1

Step 0

Start a new chat off your main account

Step 1

Open the Virelia Star System sample notes.

Step 2

Paste the Virelia notes and your Version 1 study-assistant instructions in the same message, the prompt should use:

Goal + Context + Rules + Output

Step 3

Add to the end of the prompt: Reply only with “ready”

Step 4

Send the prompt

Your assistant should be able to:

- Explain concepts

- Quiz you

- Create flashcards



Test Version 1

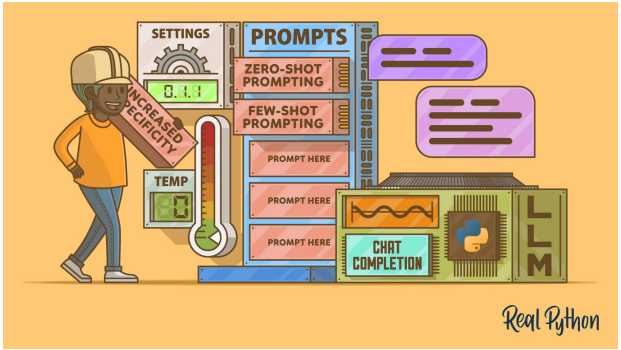
Keep using the same chat

Test 1, Explanation: Explain the event horizon of Nox like I'm a beginner

Test 2, Quiz: Quiz me on the Virelia Star System (answer 1 quiz question)

Test 3, Flashcards: Create 5 flashcards from these notes

- Did it follow your instructions?
- Did you like the amount of detail?
- Did the quiz behave how you wanted?
- Were the flashcards actually useful?



Test the limits of your assistant

Can your assistant handle missing info? Ask these questions **w/o** changing your prompt

- How many moons does Asteron have?
- How long does Belara take to orbit Virelia?
- What is the mass of the Nox black hole?

What does it do?

- Recognize info is missing?
- Makes an unsupported assumption
- Invents a specific answer?

Hallucinations

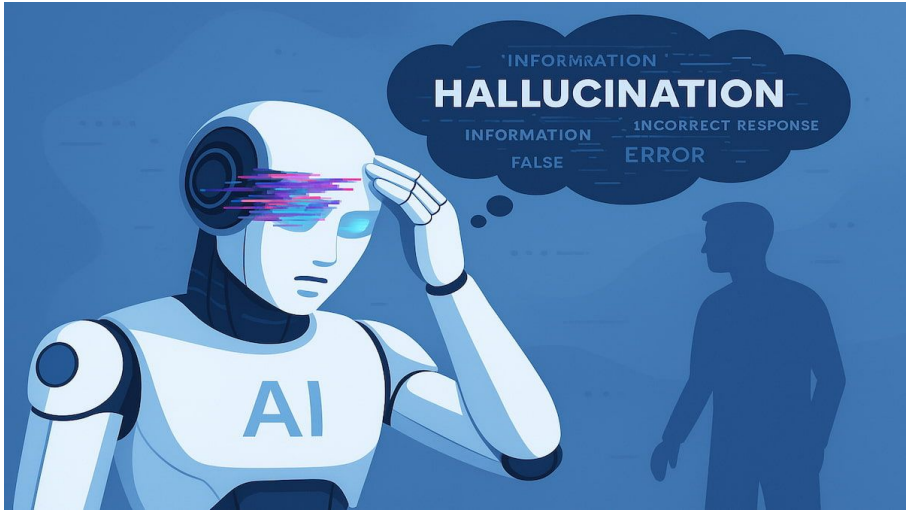
An AI hallucination occurs when a **model generates information that is false, fabricated, or unsupported.**

Hypothetical AI Response: Nox has a mass of 14.7 solar masses.

This information was never provided.

Confident ≠ Correct

Is every bad response a hallucination?



Not every bad response is a hallucination

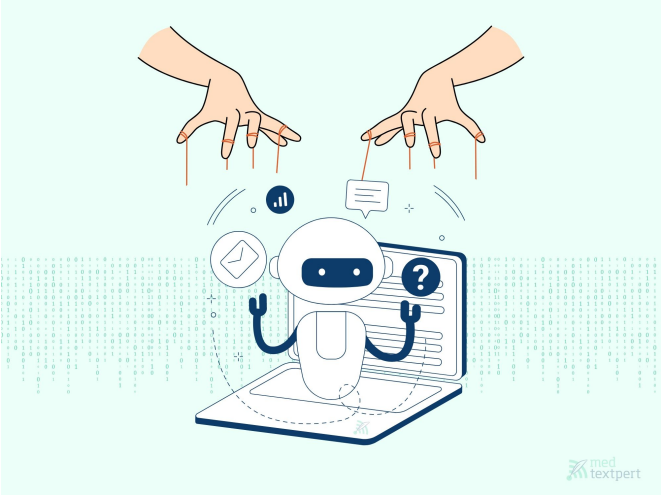
AI can fail in different ways:

- Accuracy: is the info correct?
- Grounding: is the answer supported by the source?
- Instruction-following: did it obey your rules?
- Format: did it respond the way you requested?

Evaluate before you trust

Before accepting an AI Response, ask yourself:

1. Accurate? Does it match info I know or provided?
2. Grounded? Can I trace the answer back to my source?
3. Relevant? Did it answer what I actually asked?
4. Following instructions? Did it follow the rules I gave it?
5. Useful? Does this response actually help me?



Activity 2: Improve Your Prompt

Use what you discovered

1. Identify a behavior to improve
2. Change the relevant instruction
3. Open a new tab (don’t close the old one!)
4. Create new chat
5. Paste same notes and add your version 2 prompt (again add at the end: reply only with “ready”)
6. Rerun the same tests

Previous Prompts

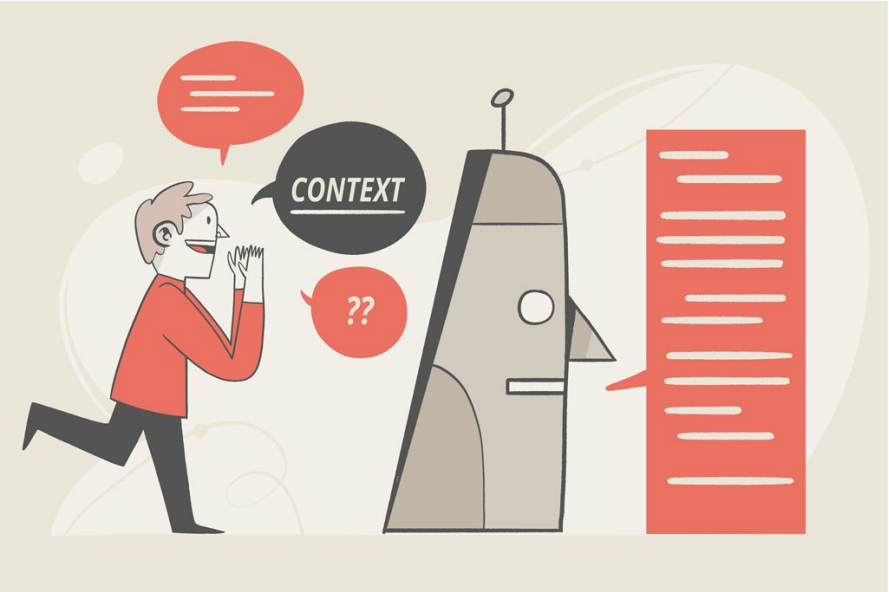
1. Explain the event horizon of Nox like I’m a beginner
2. Quiz me on the Virelia Star System (answer 1 quiz question)
3. Create 5 flashcards from these notes
4. How many moons does Asteron have?
5. How long does Belara take to orbit Virelia?
6. What is the mass of the Nox black hole?

Version 1 vs Version 2

Did your assistant improve?

Observe behavior -> Identify what to improve -> Revise the relevant instruction -> Test again

Prompting is iterative



Example of a strong study assistant prompt

Goal: You are my study assistant. Help me learn from the course notes I provide.

Context: Treat the provided notes as the source material for our study session.

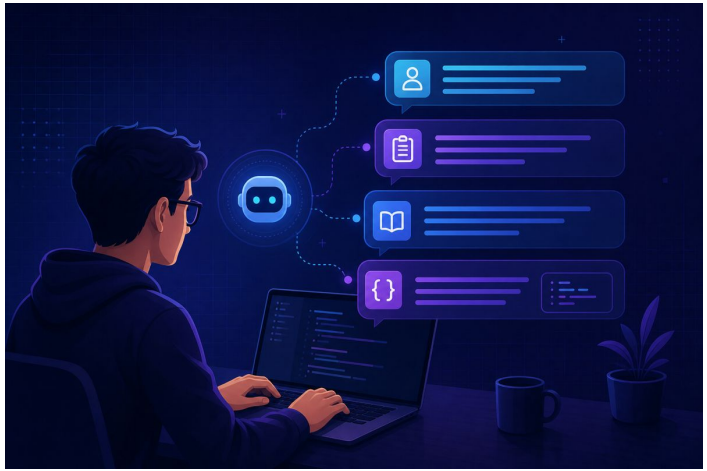
Rules: Use only information supported by the provided notes. If the notes don't contain enough information, say so instead of guessing or inventing details. When quizzing me, ask one question at a time and wait for my response before giving feedback.

Output: Use clear, beginner-friendly explanations. Keep flashcards concise. When I answer a quiz question, briefly explain why my answer is correct or incorrect.

Prompting isn’t about specific phrases or words

Good prompting is not finding a secret phrase that makes the AI perfect

Good prompting IS defining, testing, evaluating, and improving



Use AI to Support Learning

Good uses

- Ask for explanations
- Practice concepts
- Generate study questions
- Get feedback on understanding

Remember

- Follow your professor/course AI policy
- Verify AI-generated info

Save your prompt

Today you designed the **behavior**

Workshop 2:

You -> Python Application -> AI Model

The application will handle those instructions for you



Workshop #3 Preview

Today our notes were tiny, but what if we needed to feed the AI assistant:

- 20 lectures
- 300 pgs of notes
- An entire semester of material

Workshop 3 will make our application retrieve relevant notes before sending them to the AI model (RAG: Retrieval-Augmented Generation)

What You Learned

Prompt Design

- Goal
- Context
- Rules
- Output

AI Evaluation

- Grounding
- Hallucinations
- Instruction-following

Process: Define -> Test -> Evaluate -> Improve







Stay connected with BearAI! Website: bearai-bu.github.io